Our Journey Through Design Thinking

From a broad concern about AI to TrustLayer

Our Design Thinking journey began as a messy, open-ended exploration of a very current problem: how people use AI when they do not fully know whether they can trust it.

At the start, our topic felt huge. AI is everywhere: in assignments, work, healthcare, decision-making, writing, research, and everyday problem-solving. Everyone had an opinion about it, but the problem was not immediately clear. Some people used AI daily. Some avoided it. Some trusted it too much. Some did not trust it at all. Our first challenge was to move from a broad theme into a specific human problem.

Over the course, our project evolved through theme selection, mentor discussions, surveys, personas, empathy mapping, root cause analysis, problem statements, ideation methods, poster work, website documentation, and finally our solution: **TrustLayer**, a tool designed to make AI-generated answers more transparent, checkable, and responsible.

This is the story of how we got there.

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1. Starting point: choosing a theme and forming direction

Our group began by discussing possible themes and ranking them based on interest and relevance. From the beginning, the theme that stood out to us was connected to **digital integrity, intellectual ethics, and the responsible use of AI**.

At first, the theme felt broad and abstract. We knew AI was creating new problems, especially around truth, trust, academic honesty, privacy, and dependence. But we did not yet know which user group we were solving for or what the exact pain point was.

The early phase of the course was about orientation. We formed the team, looked at the course resources, discussed mentor expectations, and tried to understand what Design Thinking expected from us. We were introduced to ideas like wicked problems, user-centered thinking, and the importance of documenting not only the final solution but also the process behind it.

A major realization from this stage was that our project could not simply say, “AI has risks.” That was too generic. We needed to understand how real people experience those risks in specific situations.

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2. Understanding the problem space: AI is useful, but trust is unclear

Our first discussions around AI showed that people were not simply divided into “AI users” and “AI non-users.” The reality was more complicated.

Many users found AI helpful because it saves time, generates ideas, explains concepts, summarizes content, and supports productivity. But the same users also hesitated when the output mattered. They worried about wrong answers, hallucinations, lack of sources, unclear boundaries, and the ethical question of when AI use becomes irresponsible.

This became one of our earliest insights:

> People are not confused about whether AI is useful. They are confused about when AI is safe, appropriate, and trustworthy enough to rely on.

This shifted our thinking. The problem was not just adoption. It was not just misuse. It was the uncertainty between usefulness and responsibility.

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3. Empathy work: surveys, user experiences, and early research

To avoid designing from assumptions, we moved into empathy work. We created survey material to identify the groups and individuals affected by AI tools and to understand how different users interact with AI.

The survey helped us look at:

- how frequently people use AI tools,
- what they use AI for,
- where they hesitate,
- what makes them trust or distrust AI,
- whether they verify AI-generated answers,
- and what kind of support would make AI use feel safer.

Alongside the survey, we discussed real experiences with AI. These conversations helped us see that AI trust depends heavily on context. A student using AI to understand a topic has a different risk level from a doctor using AI to support a medical decision. A professional drafting content has different concerns from someone using AI casually for quick answers.

This empathy stage helped us move away from one generic user and toward multiple user types.

4. Persona development: seeing the problem through different users

From the research and discussions, we developed four major personas. Each persona represented a different relationship with AI trust.

Persona 1: The Conflicted Power User

This persona represents students or frequent AI users who rely on AI often but feel conflicted about it. They use AI for learning, writing, brainstorming, or completing tasks, but they are not always sure whether they are using it responsibly.

Their tension is internal: AI helps them move faster, but it also creates doubt about originality, learning, and academic integrity.

Persona 2: The Cautious Professional

This persona represents working professionals who see AI as useful but cannot blindly rely on it. They may use AI to draft, summarize, or organize information, but they still need accuracy, context, and confidence before using the output.

Their problem is not lack of interest. Their problem is that verification takes time, and without clear signals, AI feels like partial help rather than a complete productivity gain.

Persona 3: The High-Stakes Skeptic

This persona represents people in fields where mistakes can have serious consequences, such as healthcare, law, finance, or other accountability-heavy domains.

For them, trust is not optional. They need reliability, explainability, and clear boundaries. If AI cannot show uncertainty, sources, and accountability, they would rather avoid it.

Persona 4: The Low-Trust Casual User

This persona represents everyday users who may use AI casually but do not fully believe its answers. They may ask AI questions, but they still feel the need to cross-check everything because the answer does not show where it came from or how confident it is.

Their problem is not technical complexity. Their problem is simple: AI answers often feel too fluent to question but too unsupported to trust.

Developing these personas was a turning point. It showed us that “AI trust” is not one problem. It is a family of problems shaped by user goals, stakes, confidence, and consequences.

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5. Root cause analysis: moving beyond surface-level symptoms

After building personas, we moved into root cause analysis. Initially, it was easy to say that users do not trust AI because AI can be wrong. But that explanation was too shallow.

Using the 5 Whys and root cause maps, we dug deeper.

We asked:

- Why do users hesitate to trust AI?
- Why do they feel responsible for verifying everything themselves?
- Why do AI outputs feel risky even when they sound confident?
- Why do users not know when AI should or should not be used?
- Why does AI fail differently for different personas?

Through this stage, we realized that the deeper issue was not only inaccuracy. It was the absence of visible trust signals.

AI tools often provide polished answers without showing:

- confidence level,
- uncertainty,
- source quality,
- factual support,
- ethical limitations,
- risk level,
- or whether human judgment is required.

This created a major insight:

> Users carry the burden of uncertainty because AI gives answers without enough context to judge them.

This became the foundation for our final problem framing.

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6. Problem statement evolution

Our problem statements evolved persona by persona.

For students, the issue was responsible use and learning integrity. For professionals, the issue was time lost in verification. For high-stakes users, the issue was reliability and accountability. For casual users, the issue was lack of clear signals about accuracy and sources.

Across all personas, the common thread was this:

> Users need a way to understand how much weight to give an AI-generated answer before they act on it.

This reframing was important because it transformed our project from a broad awareness campaign into a design opportunity. We were no longer trying to "make people trust AI." Instead, we were trying to help people **calibrate trust**.

That distinction mattered.

Blind trust is dangerous. Total distrust is limiting. Calibrated trust is useful.

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7. Ideation: from many possibilities to one focused solution

Once the problem was clearer, we began ideation. We generated How Might We questions for each persona, such as:

- How might we help users understand when an AI answer is reliable?
- How might we surface uncertainty without interrupting the user's workflow?
- How might we help students use AI without losing learning integrity?
- How might we help professionals verify outputs faster?
- How might we make AI reasoning more inspectable instead of treating it like a black box?
- How might we give users visible anchors for sources, confidence, and limitations?

We also used structured ideation methods like SCAMPER and TRIZ. These helped us move beyond obvious ideas and think about the problem from different angles: replacing, combining, adapting, reversing, or simplifying parts of the AI interaction.

During this stage, several possible solution directions emerged. Some focused on education. Some focused on warnings. Some focused on verification. Some focused on interface design.

We evaluated ideas based on desirability, feasibility, and viability. The strongest direction was a lightweight trust layer that could sit between the user and AI-generated output.

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8. Final concept: TrustLayer

Our solution became **TrustLayer**.

TrustLayer is a tool that analyzes AI-generated answers and helps users judge whether the output is reliable enough to use. Instead of replacing the AI tool, it adds a transparent verification layer on top of it.

The idea is simple: when a user receives an AI-generated answer, they should not be left alone to guess whether it is safe. TrustLayer gives visible signals that help them understand the answer’s trustworthiness.

TrustLayer focuses on features such as:

- confidence scoring,
- source and claim checking,
- uncertainty flags,
- risk indicators,
- limitation warnings,
- and guidance on whether human verification is needed.

The goal is not to make users blindly trust AI. The goal is to help users decide how much trust is appropriate in a specific context.

In that sense, TrustLayer shifts AI from a black-box answer generator into a more transparent and accountable assistant.

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9. Prototype and documentation work

After finalizing the concept, we moved toward visible deliverables: the poster, the website, and the prototype/documentation.

The poster helped us compress the project into a clear design story: user problem, insights, How Might We questions, top ideas, and final solution. It forced us to decide what the core message really was.

The website became a place to document the journey and show the project in a structured way. It was not just about presenting the final tool; it was about showing how the solution emerged from research, empathy, root cause analysis, and iteration.

This stage also made us realize that the story of our process was as important as the final product. The final feature matters, but the journey shows why that feature exists.

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10. What changed in our thinking

At the beginning, we thought the problem was something like:

> People do not trust AI.

By the end, our understanding became much sharper:

> People use AI even when they are unsure about it, but they lack contextual signals to judge whether an output is safe, accurate, ethical, or appropriate to rely on.

This change in framing was the biggest learning from the course.

We learned that Design Thinking is not just about creating a solution. It is about repeatedly asking whether we are solving the right problem.

Each stage changed the project:

- Theme selection gave us direction.
- Surveys gave us real user input.
- Personas showed us different types of AI trust problems.
- Root cause analysis pushed us below surface-level assumptions.
- Problem statements clarified the real pain points.
- HMW questions opened solution possibilities.
- Ideation methods helped us explore alternatives.
- The poster and website forced us to communicate clearly.
- TrustLayer emerged as a focused response to everything we learned.

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11. Final reflection

Our Design Thinking journey was not linear. We did not start with TrustLayer. We started with a broad concern about AI, confusion about the exact problem, scattered user experiences, and multiple possible directions.

Through the course, we learned to narrow the problem without oversimplifying it. We learned that AI trust is not a yes-or-no issue. It depends on the user, the task, the consequences, and the level of support provided by the system.

TrustLayer is our response to that insight.

It is built around the belief that the future of AI should not depend on blind faith or total rejection. Users need tools that make AI outputs easier to inspect, verify, and responsibly act upon.

Our journey through this course helped us move from asking:

> “Can people trust AI?”

To asking:

> “How can we help people know when, where, and how much to trust AI?”

That shift is what shaped our final solution.